Inferential Statistics 4 Aug

Tuesday, 4 August 2026 06:58

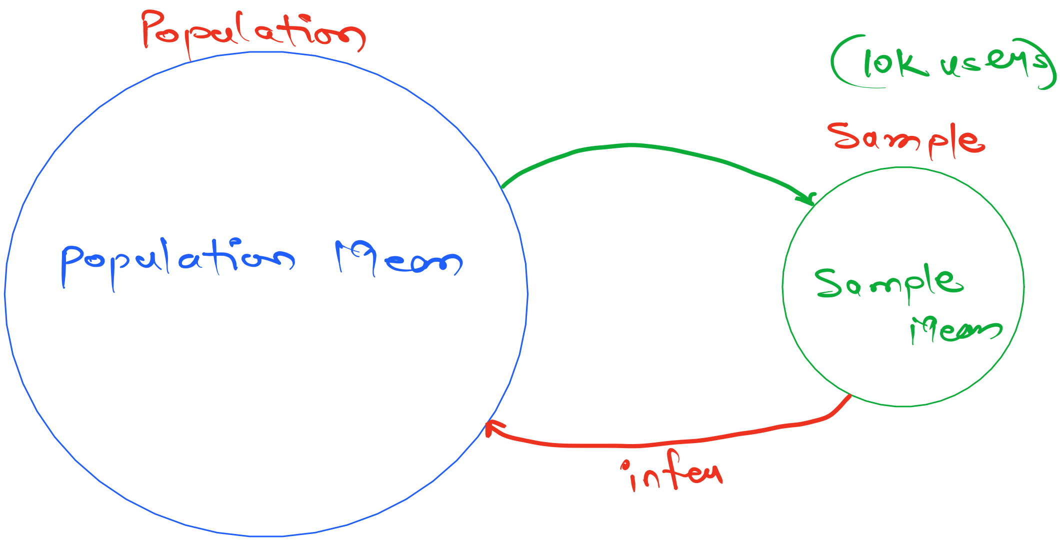
There are 2 branches in statistics:

(1) Descriptive statistics

(2) Inferential Statistics ←

- → Population & Sample
- → Confidence Interval
- → Hypothesis Testing

Inferential statistics



Youtube: Avg daily watchtime of users:

Population: All users of Youtube.

(2 billion)

(Pop size)

Population Data : Daily watchtime of every user in the population

1. 83 mins
2. 97 mins
3. 191 mins

⋮

(2B) 200 mins

↓

Pop Mean = 195 mins / day

→ Time Constraint

→ Budget Constraint

What are the ways to estimate population mean using Sample Mean ?

1. Point Estimation

Point Estimate

$$\boxed{\text{Population Mean} = \text{Sample Mean}}$$

2. Confidence Interval :

Confidence Interval = Point Estimate ± Margin of Error

S_1
↓
185 mins/day

= Sample Mean ± MOE

No

= 185 mins/day ± 5 mins/day

CI = 180 mins/day to 190 mins/day

interval

Avg Daily watchtime of users is b/w 180 mins/day to 190 mins/day.

Prob:

Seagate/WD/
Sandisk

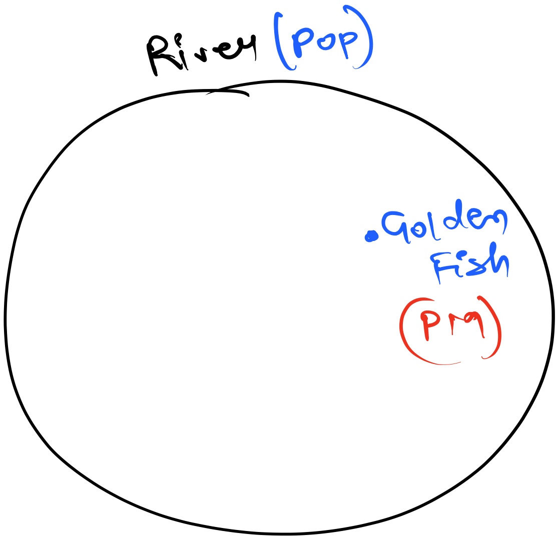
HDD/SSD

How much should be the optimal warranty given to the customers?

Pop: All HDD manufactured by them.
(200k)

Why it is called confidence Interval not just Interval?

Confidence Level	Interval
90%	185 mins ± 5 mins
95%	185 mins ± 7 mins
99%	185 mins ± 10 mins



~~Hook~~

Fishing Net (CI)

Small

Med

Large

Formula to calculate confidence Interval:

$$CI = SM \pm MOE$$

$$= SM \pm \text{critical value} * \text{Standard Error}$$

$$= \bar{x} \pm Z^* * \frac{S}{\sqrt{n}}$$

$\bar{x}$: Sample Mean

Z^*: critical value

S: standard deviation of the sample

n: Sample size

Z^*	CL
1.65	90%
1.96	95%
2.58	99%

Z-table